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Magnetic polarity stratigraphy of a Plio-Pleistocene marine sequence of North Island, New Zealand

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Thick sequences of relatively undisturbed Plio-Pleistocene sediments in the Wanganui Basin, North Island, New Zealand consist of well exposed silts, clayey silts, sandstones, rare limestones, and several tephras Oriented specimens were collected from a section more than 2500 m thick and palaeomagnetic measurements were made using A C demagnetisations in fields up to 35 mT With the aid of tephrochronology the age of the upper sequence is now well established and falls within the Matuyama epoch The lower two-thirds of the section except for the basal 500 m is predominantly normally magnetised and is interpreted as a very extended sequence of the Gauss epoch The lowest 500 m then represents the Gilbert epoch The Plio-Pleistocene boundary, as defined at Vrica, Italy, falls within the upper part of the section studied, in the Upper Nukumaruan stage For the first time a reliable correlation is made with the international boundary, using as intermediaries the palaeomagnetic and tephrostratigraphy of deep-sea cores from the southwest Pacific

As a result of the high deposition rate (of the order of 1 2 m/ky) and the apparent lack of unconformities, the temporal resolution is high, short-lived magnetic events are detected, especially in the lower Matuyama and upper Gauss epochs These generally correlate well with events reported from other extended sections

1. Introduction

Few sedimentary sequences exist through the Upper Pliocene/Pleistocene that are thick, complete and tectonically uncomplicated. Such an extended sequence is exposed in North Island, New Zealand and is theoretically ideally suitable for paleomagnetic studies. The longest complete sedimentary sequences of this time range so far studied are those of Liddicoat et al. [1] with a 930 m thick core of lacustrine and alluvial material through Searles Lake, U.S.A and of Cooke et al [2] and Rońai [3] with two boreholes, 1200 m thick, in the non-marine sequence of the Great Hungarian Plain. The Searles Lake core was interpreted as extending through the Brushes epoch almost to the base of the Gauss. Five zones of inclination were recognised that are not on the standard polarity time scales. In any extended sequence this must be expected as most paleomagnetic stratigraphy has been based on temporally concentrated sediments in the deep sea or on lava sequences where time is erratically represented The Hungarian sequences also confirm the generally accepted time scale with the presence of fine detail not usually found.

The section studied in New Zealand has the added advantage of the presence of tephra horizons which either have been dated radiometrically or can be correlated to deep sea equivalents with age interpolations from paleomagnetic stratigraphy.

2. Geological setting

Thick deposits of Pliocene and Pleistocene marine sediments are exposed in the Wanganui Basin of North Island, New Zealand (Fig. 1). This marine sequence is thickest (approximately 4360 m) in the geographical centre of the basin, and is generally continuous with few structural complications. The sediments lie unconformably on pre-Cretaceous greywackes.

In the basin centre, i.e. in the region of the Rangitikei River (Fig. 1), the beds dip generally to the south and southwest between 6 and 10°. The

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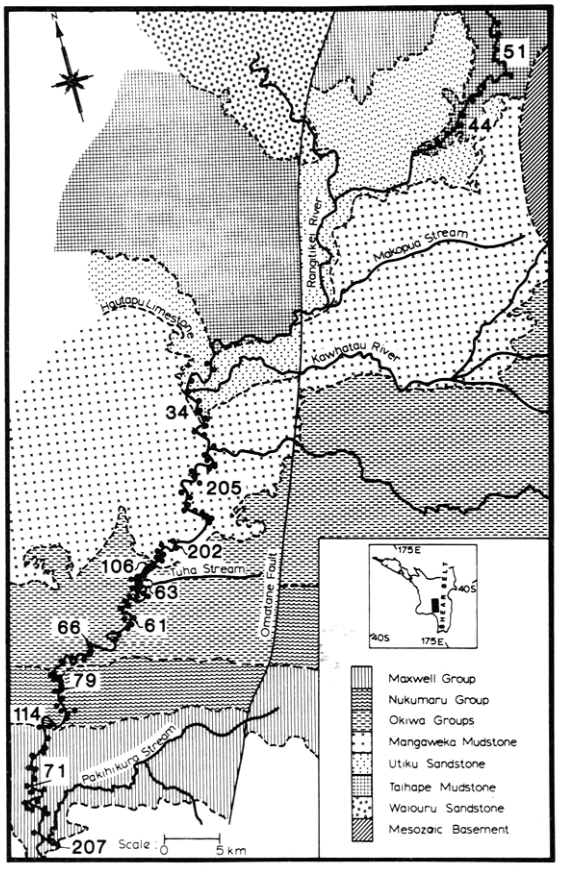


Fig 1 Geological map of the area studied Every tenth sample site is numbered Inset shows position of the North Island Shear Belt

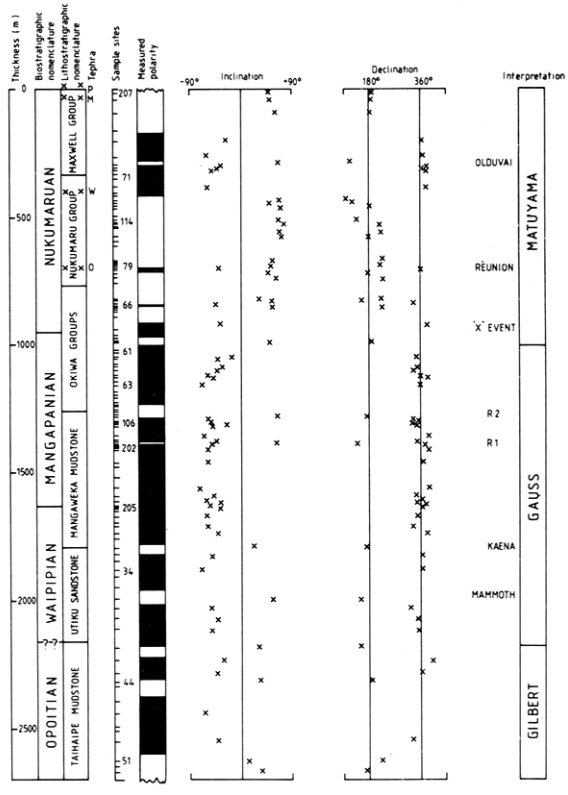


Fig 2 Bio, litho-, and magnetostratigraphy of the section studied For position of stage boundaries, see text *P*, *M*, *W* and *O* are the positions of the Pakihikura Pumice, Mangahou Ash, Waipuru Ash and Ohingaiti Ash Every tenth sample site is numbered

drainage pattern across the basin is also to the south, the rivers flowing down dip from the greywacke basement through the complete sedimentary sequence of Pleistocene and Upper Pliocene sediments.

The complete sequence in the Rangitikei Valley was not sampled because (a) the lowermost formation, the Waiouru Sandstone is inaccessible, and (b) the upper portion is composed of generally coarser material, often volcaniclastic and frequently iron stained. Thus the 2600 m studied here represents the large middle sequence of Plio-Pleistocene sediments. The lower half of the sampled sequence (Fig. 2) is predominantly mudstones with occasional shell limestones, conglomerates and upwards increasing numbers of concretionary horizons. A major limestone, the Hautapu Shell Limestone, lies between the Mungaweka Mudstone and the Utiku Sandstone (Fig. 1). The upper section studied is composed predominantly of sandy mudstones which are fossiliferous in many places. Three tephra beds, Ohingaiti, Waipuru and Mangahou Ashes, which are critical to the palaeomagnetic interpretations, outcrop in this sequence. Above the Maxwell Group (Figs. 1 and 2) there is a dramatic lithological change as a massive influx of volcanic detritus from the Central Volcanic District to the north of the basin is recorded in the marine sediments. The ash sequence has been dated by the fission track method [4] and provides a baseline for palaeomagnetic studies. The age at the top of the sequence in the current study is 1.1 Ma [5], a fission track age on the Pakihikura Pumice (Fig 2). The Brunhes/Matuyama boundary has been tentatively identified by Seward [6] 400 m above

the top of the section studied here.

The biostratigraphy of the sequence is discussed by Superior [7], Te Punga [8] and Collen [9]. Stage boundaries in the Rangitikei sequence are apparently difficult to assign as many important species are not present (N. de B. Hornibrook, personal communication, 1983) so all estimates must be considered broad. Hornibrook (personal communication, 1979) places the Opoitian/Waipipian boundary at approximately the top of the Taihape Mudstone, and the Waipipian/Mangapanian boundary in the lower third of the Mangaweka Mudstone (Fig. 2). The Mangapanian/Nukumaruan boundary is that defined by Fleming [10]. Because of the paucity of planktonic foraminifera, the Plio-Pleistocene boundary may still be controversial but according to Beu and Edwards [11] and Edwards [12], the best biostratigraphical estimate of the boundary as defined at Vrica is now in the upper part of the Nukumaruan Stage. One of the aims of the present study was to more closely position the internationally defined Plio-Pleistocene boundary within the New Zealand stages.

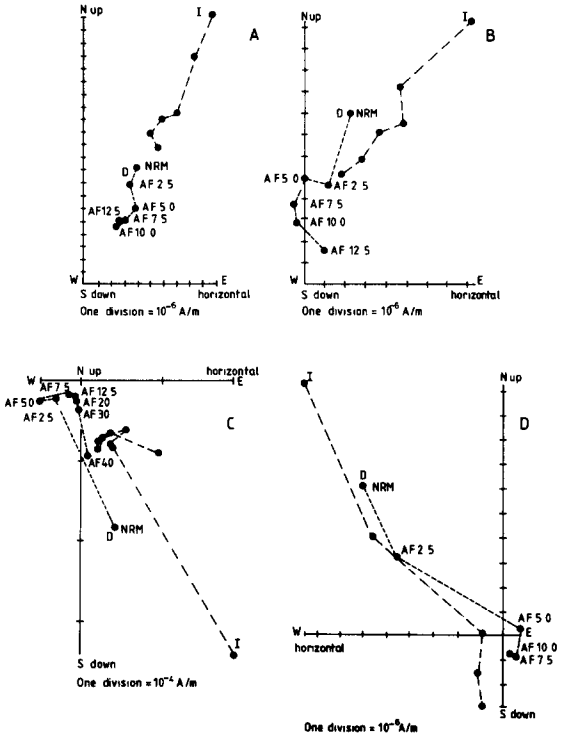


Fig 3 Modified Zijderveld plots for sediments in the sequence (A) is typical of the Olduvai sequence, (B) is in the top of the Gauss, (C) and (D) are both in the upper reversed section

3. Sampling and laboratory procedures

The river was accessed by jet boat, and the samples were cored from river cliffs. Samples were generally taken only in blue grey mudstones. At least three specimens were drilled per site with less than 10 cm stratigraphic difference between cores. Initial samples were taken at stratigraphic intervals of 50 m or less; a second collection was made later (in sequences that were difficult to interpret) with sites as close as 10 m apart.

In previous work on sediments in an adjoining basin, Lienert et al. [13] used thermal cleaning throughout. It was reasonably satisfactory except that chemical alteration fairly clearly appeared at temperatures between 250° and 350°C Allowing for this restriction with chemical cleaning, there appears little difference between results achieved with A C or thermal cleaning. A.C. cleaning was more convenient in the present study and was therefore used. The point of stability for each core was individually assessed, by cleaning in stepwise increments to 70 mT Because of the generally low intensity of magnetisation of the sediment, measurements at higher demagnetisation fields were approaching the noise level of the system. In most cases this point was less than 35 mT (Fig. 3). Directions of magnetisation were measured in a cryogenic magnetometer

4. Results

4 1 General

The NRM values for all measurements (Fig. 4A) show a broad distribution of directions for the lower hemisphere (reversed). Their mean declination is 062°, inclination 79° with a 95% cone of confidence of ±19°. A low Fisher k value of 2 means that little confidence can be placed on these statistics. The cores with upper hemisphere (normal) inclination have magnetisation directions clustering around the present magnetic field vector. The mean direction has declination 022°, inclination −60° with a 95% cone of confidence of ±5°. The present magnetic field direction at the collecting sites is 022° declination, −59° inclination.

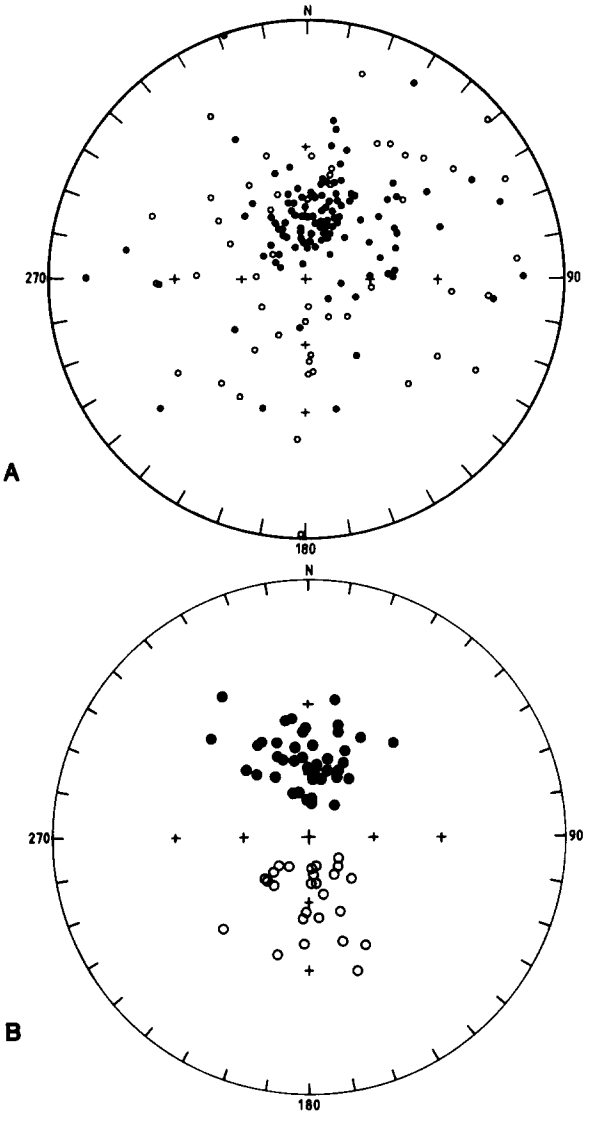


Fig 4 A Stereonet projections of all NRM directions corrected for tilt Solid symbols indicate upper hemisphere projections, (normal) open indicate lower hemisphere projections (reversed) Mean directions upper hemisphere, $D = 022°$, $I = -60°$, $\alpha_{95} = \pm 5°$, lower hemisphere, $D = 062°$, $I = 79°$, $\alpha_{95} = \pm 19°$ B Stereonet projections of statistically significant site means Symbols as above Mean directions upper hemisphere, $D = 3°$, $I = -53°$, $\alpha_{95} = \pm 5°$, lower hemisphere, $D = 179°$, $I = 65°$, $\alpha_{95} = \pm 8°$

After cleaning and selection of end points from modified Zijderveld diagrams (Fig. 3) and stereoplots (Fig. 4), the data fell into time sequential groups showing clearly normal and reversed behaviour (Fig. 2) Some sets of stratigraphically consecutive measurements satisfied the stability criteria but gave generally intermediate geomagnetic pole positions, e.g., at 400 m, just beneath the Waipuru Ash. We believe they represent periods of genuine magnetic transition but because of their intermediate nature they are not plotted in Fig 2 They are to be the basis of a further more detailed study.

The stereographic projections of the consecutively consistent normal and reversed directions are shown in Fig. 4B. After tilt correction (6–10°) the normal (upper hemisphere) mean declination is 3°, mean inclination −53°, with a 95% cone of confidence of ±5°. For the reversed (lower hemisphere) the mean declination is 179°, mean inclination 65°, with a 95% cone of confidence of ±8° (Fig 4B) From the plots we note the following points:

(1) Within our confidence limits normal and reversed are antipodal

(2) The normal directions of the cleaned samples do not overlap those of the normal NRM directions at the 95% level; neither do they overlap the present field direction, (022°, −59°) indicating little influence of current magnetic field. This result adds confidence to our cleaning technique.

(3) Within the confidence limits there is no offset of the mean magnetic meridian from the geographic meridian over a time period of approximately 3 Ma. This contrasts with the mean direction found by Walcott et al. [14] for measurements on samples of slightly older age (5–6 Ma) some 100 km away, but east of the North Island Shear Belt (inset, Fig. 1). These results fit in well with the current tectonic model for plate motions for the New Zealand region (H.W. Wellman, personal communication, 1986) From plate tectonics the declination change for the sediments in the Rangitikei section should be only of the order of 2–4°

4.2. Stratigraphic interpretation of results

Interpretation of any paleomagnetic sequence usually depends on some other well founded age data within the section, be it palaeontological, radiometric, etc. In this case, the top of the section has two dated tephras The Pakihikura Pumice (Fig 2) has a glass fission track date of 1 06 ± 0.32 Ma [4] which was confirmed by a zircon fission track age of 1.06 ± 0.36 Ma [5]. (All errors are quoted at the 2σ level.) The Mangahou Ash, 20 m

below, is dated on glass at 1.26 ± 0.34 Ma. These two ages are stratigraphically consistent with each other as they are with younger ashes in the sequence [4,6].

These dates place the top of the sequence within the Matuyama epoch. Further, the Brunhes/Matuyama boundary has been tentatively located by Seward [6] at a stratigraphically higher point between two tephras dated at 0.74 ± 0.18 Ma and 0.61 ± 0.12 Ma.

4 3. Matuyama epoch

On the basis of dominantly reversed pole positions and the fission track dates on the tephras, it is concluded that the mainly reversed sequence from 0 to 1000 m is part of the Matuyama epoch. Within it is a normally magnetised sequence from 200 to 400 m. We interpret this as the Olduvai event with ages of 1.67–1.87 Ma assigned to it by Mankinen and Dalrymple [15]. Immediately below the normal interval, the directions from five closely spaced consecutive sites are intermediate, and they have been rejected from our column. Data for the Réunion event are sparse but we suggest that it occurs just below the Ohingaiti Ash (Fig. 2). A possible normal polarity event identified as anomaly X by Heirtzler et al. [16] and Emilia and Heinrichs [17] was also tentatively recognised by Shuey et al. [18] and as N1 by Liddicoat et al. [1]. We suggest that the normal event at about 950 m in the Rangitikei Section is also the "X" event.

4.4 Gauss epoch

From 1000 to 2200 m the sediments are predominantly normally magnetised The number of events do not fit precisely with the standard reversal stratigraphy but are very similar to the results of Liddicoat et al. [1], Cooke et al. [2], and Roňai [3] (Fig 5). We take the top of the Gauss at 1000 m, and suggest that the Kaena and Mammoth events are at 1800 and 2000 m respectively although several options are obviously available. This places correlations of the event at 1250 m with the R2 of Liddicoat et al. [1] (Fig. 5), and the possible event at 1350 m may be the equivalent of their R1. The palaeomagnetic stratigraphy at Vésztö, Hungary [3], also shows two events, above the Kaena-Mammoth sequence. Interestingly, in Mankinen and Dalrymple's [15] lists of all the available dated material three reversed samples

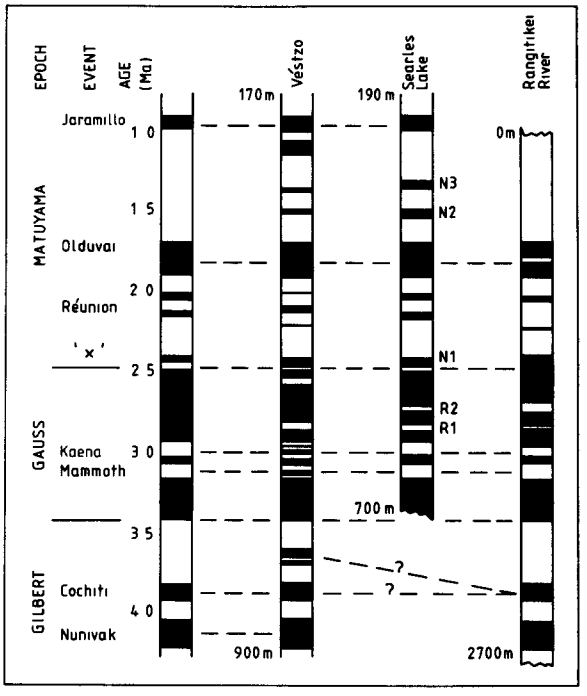


Fig 5 Correlation of the palaeomagnetic stratigraphy of sections from Europe (Vésztzo), North America (Searles Lakes) and New Zealand (Rangitikei River) compared to the reversal time scale [15] To comply with the time scale at left the columns are time not thickness related The top and bottom measurements of each column, which are not necessarily the top and bottom of each core, are included to emphasise the thickness differences between each study

fall into this interval, one at 2.70 Ma and two at 2.84 Ma.

4 5. Gilbert epoch

From 2200 m to the base of the section the sequence has several reversed and normal intervals of comparable thickness. We interpret this as being the upper part of the Gilbert epoch, but concede that closer sampling is necessary for full confirmation. This puts the age of the bottom of the section at 4.2–4.3 Ma.

5. Tephrochronology

The Waipuru Ash occurs 350 m below the top of the section. The size of the zircons (< 54 μm) in this ash are too small to be handled technically in the fission track dating process. It occurs in the sequence of predominantly normal polarity which

we have interpreted as the Olduvai event(s) (Fig. 2) In core RC12-216, taken 1600 km east of North Island, New Zealand, Watkins and Huang [19] also located a tephra (their M2) within the Matuyama epoch, in a normally magnetised sequence, interpreted as the Olduvai-Gilsa event. We conclude that the two tephra are correlatives (Fig. 6), contrary to Watkins and Huang's [19] suggestion that M2 is the distal equivalent of the Ohingaiti Ash. We now know that the Ohingaiti Ash at its type section lies within a reversed polarity sequence (Fig. 2).

In a further study, Huang et al. [20] recognised 30 dispersed volcanic ash zones in cores collected in the South Pacific east of the Balleny Islands. The dominant ash, the "Eltanin Ash" occurred in a reversed polarity sequence which was estimated to be 1.8 Ma on the basis of the Cox time scale, but is now considered to be a little older, on the basis of the Mankinen and Dalrymple [15] time scale In a recent article, Kyle and Seward [21] correlated the "Eltanin Ash" with the M2 tephra. The polarity stratigraphy now shows this not to be so—the M2 tephra is more likely to be the Waipuru Ash as discussed above, while the "Eltanin Ash" is most likely the correlative of the Ohingaiti Ash. Further, the Ohingaiti Ash is most probably the equivalent of the M1 ash (reversed) [19]. The fission track age of 1.78 ± 0.44 (2σ) on the Ohingaiti Ash, compared with the paleomagnetic ages of its possible deep sea equivalent, is within the error limits quoted

On the basis of the palaeomagnetic stratigraphy we conclude (Fig. 6) that (1) the Ohingaiti Ash is the equivalent of the "Eltanin" and M1 ashes, (2) the Waipuru Ash is the equivalent of ash C3 in the South Pacific cores and ash M2, (3) the Mangahou Ash is the equivalent of ashes D5 and M3, and (4) the Pakihikura Pumice is the equivalent of E1 and M4 ashes.

6. Plio-Pleistocene boundary

According to Tauxe et al. [22] the proposed stratotype at Vrica, Italy, occurs above the Olduvai normal event at approximately 1.6 Ma. Since 1953, the Plio-Pleistocene boundary in New Zealand has been accepted as coinciding with the base of the Nukumaruan Stage [10] because this marks the first appearance of the sub-Antarctic bivalve *Chlamys delicatula* (Hutton). But Edwards [12], using biostratigraphical correlations from New Zealand to DSDP site 284 in the Tasman Sea, and then via core V28-239 in the equatorial Pacific to Vrica, postulates the location of the Plio-Pleistocene boundary at an undefined position within the upper part of the Nukumaruan Stage. Beu and Edwards [11] estimate the boundary more precisely as lying at about the top of the Pukekiwi Shell Sand in the type Nukumaruan section, 60 km west of our Rangitikei section. Although the Pukekiwi Shell Sand does not occur in the Rangitikei section, its estimated equivalent position is 100 m below the top of our section. This is just above the top of what we interpret to be the Olduvai event. Our data thus confirm the position

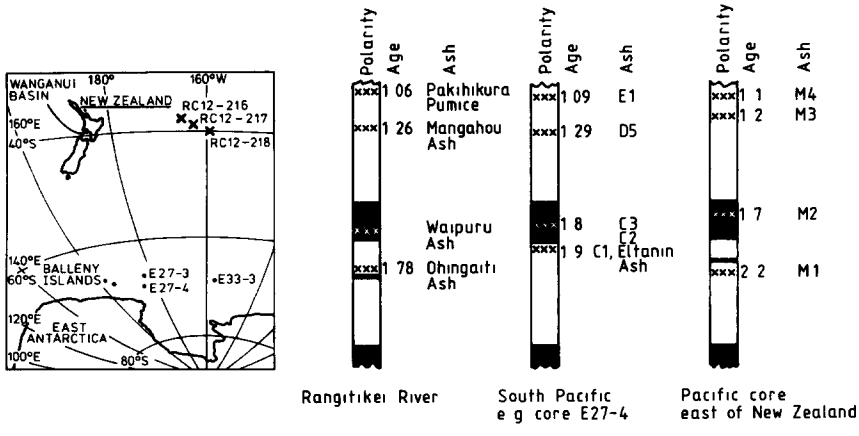


Fig 6 Tephro- and magnetostratigraphy of three widely separated localities in the South Pacific showing ash correlations

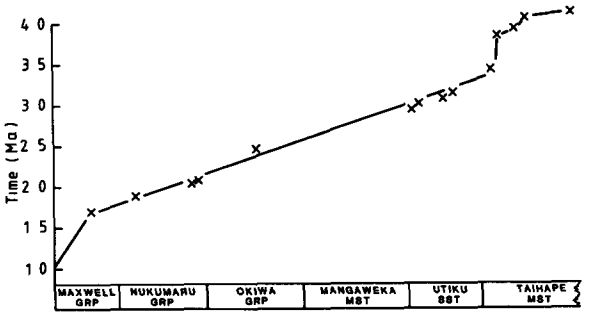


Fig 7 Sedimentation rate for the sequence studied, based on our palaeomagnetic interpretation

of the Olduvai event as estimated by Beu and Edwards. We place the Plio-Pleistocene boundary 100–150 m below the top of our section

7. Sedimentation rates

Sedimentation rates (Fig. 7) plotted on the basis of our interpretations show broadly three sections. The lowest section, predominantly the Taihape Mudstone, gives a rate of 1.7 m/ky; the broad middle portion comprising the majority of the section gives a mean rate of 1.2 m/ky. The short interval at the top of the section has a reduced rate consistent with field evidence of very shallow water sedimentation and occasional emergence at this level.

8. Conclusions

(1) In a 2.5 km thick Plio-Pleistocene section the magnetostratigraphy enables us to identify the Matuyama, Gauss and part of the younger Gilbert epochs

(2) Comparison of the magnetostratigraphy and tephrochronology with that from some deep sea cores enables us to relate the offshore tephra and dispersed ash layers to eruptive events within New Zealand.

(3) The magnetostratigraphy of our assigned Gauss epoch contains several events not recorded in the standard columns However, the pattern of events closely resembles that of measurements on other expanded sequences such as Searles Lake, California, and the Pannonian Basin, Hungary.

(4) Through the magnetostratigraphy we are able to locate the Plio-Pleistocene boundary as defined at the Vrica section. Its position is towards the top of the New Zealand Nukumaruan Stage.

(5) The resolution and continuity in this section are such that more closely spaced measurements could establish this as a type section for the Pliocene and Lower Pleistocene.

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